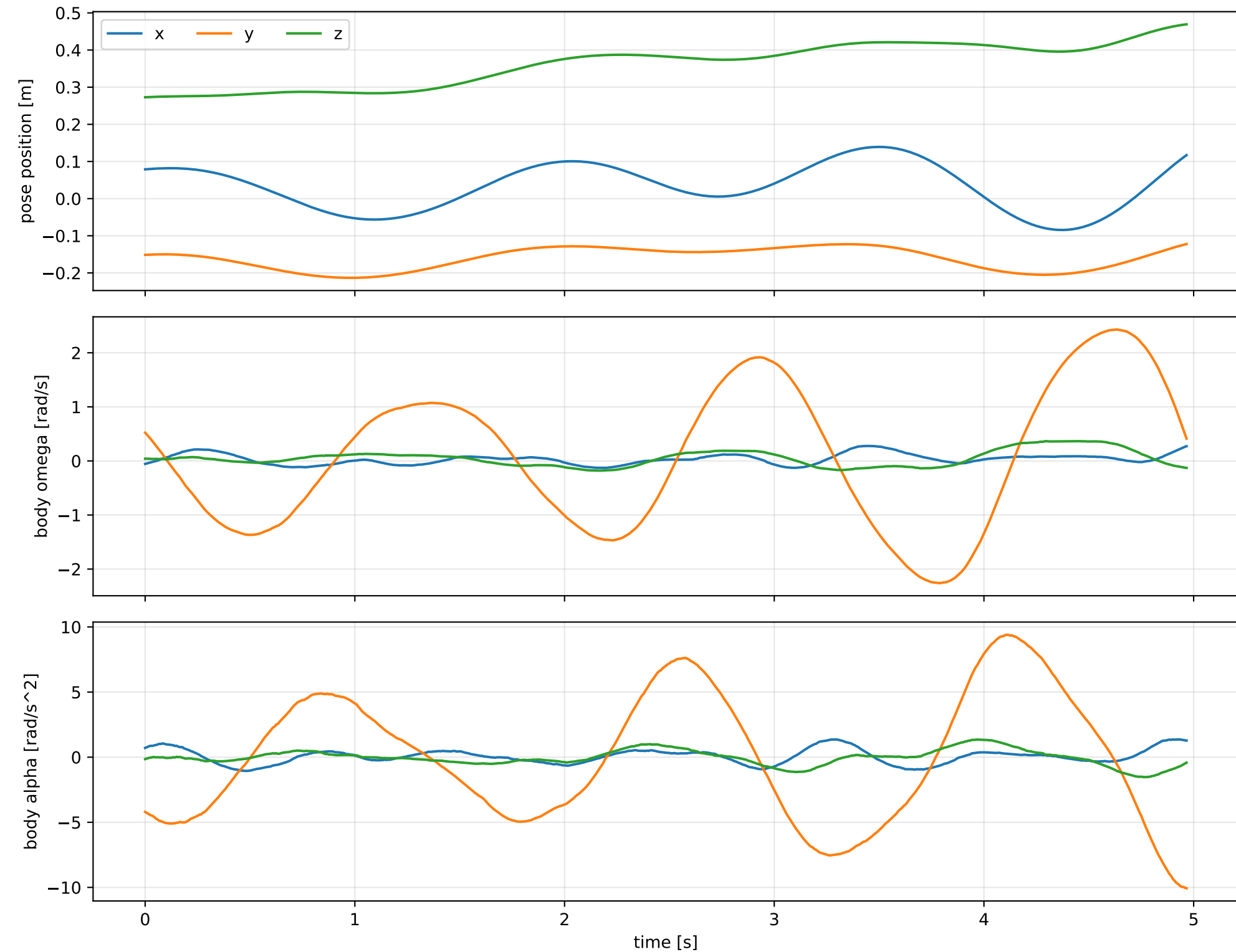
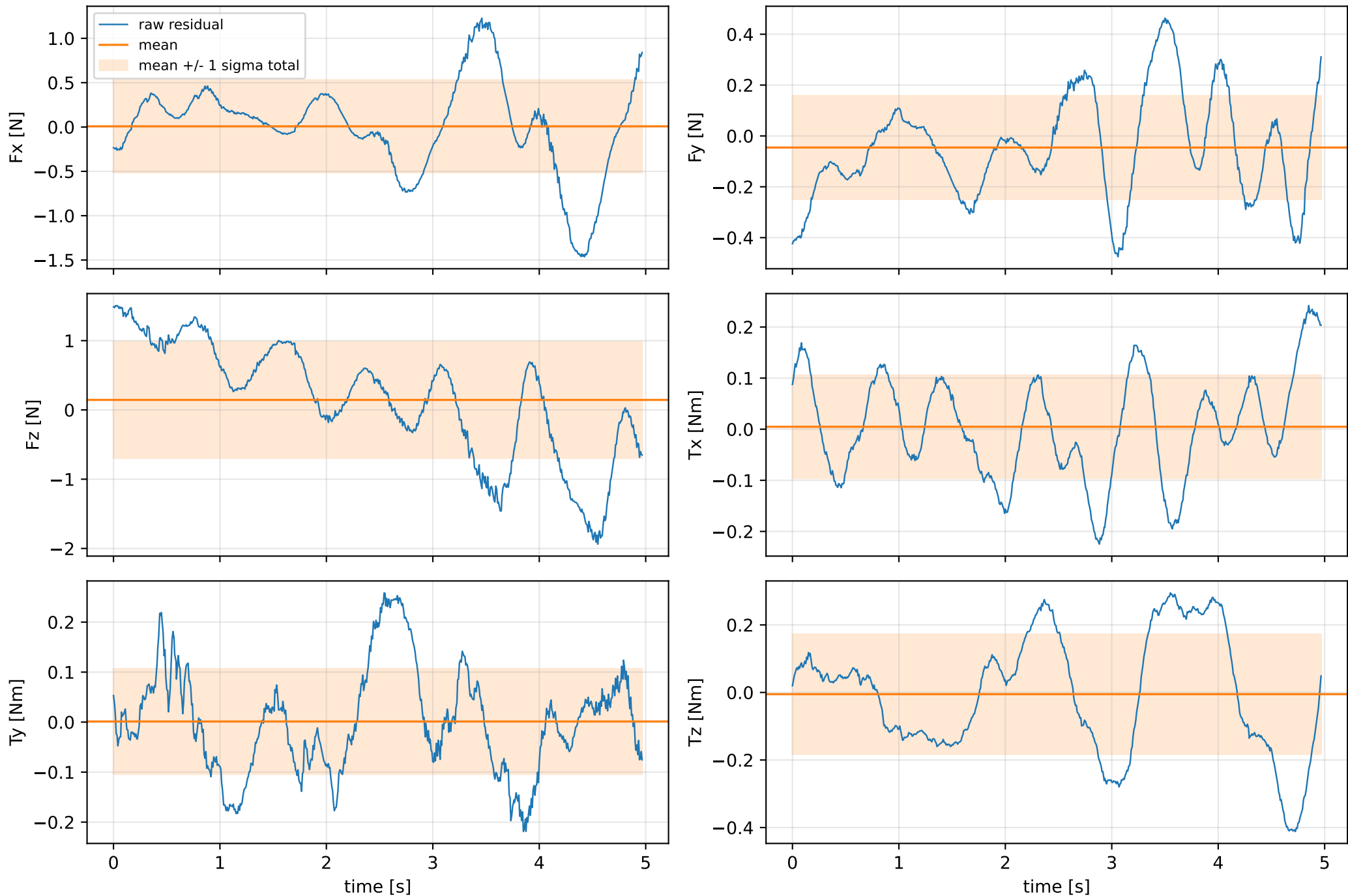


force_over_mass_rotor_2_nominal: SG trajectory and derived motion



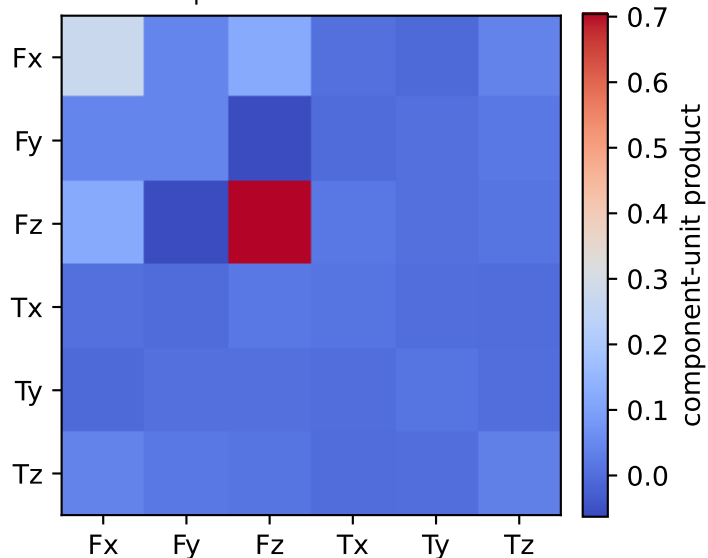
force_over_mass_rotor_2_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



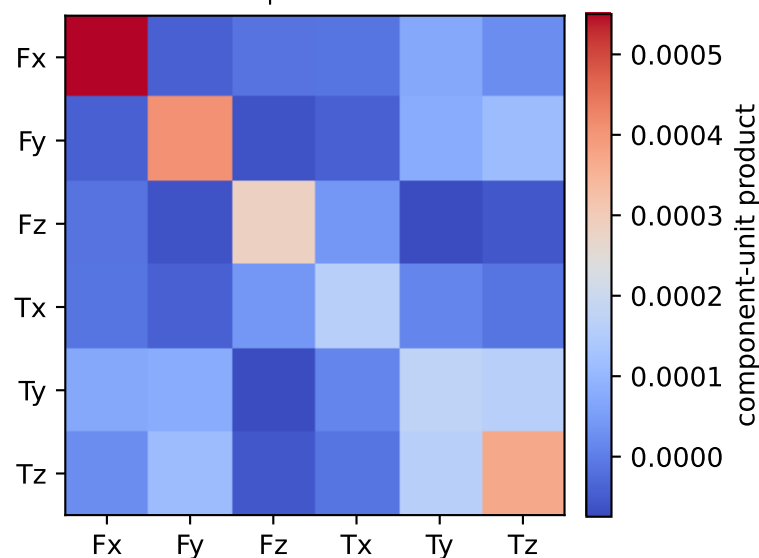
vector RMS about zero: force 1.0192 N, torque 0.22947 Nm; component std about mean: force [0.5217 0.2033 0.839], torque [0.1008 0.1052 0.1774]

force_over_mass_rotor_2_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.036 0.261 0.739]

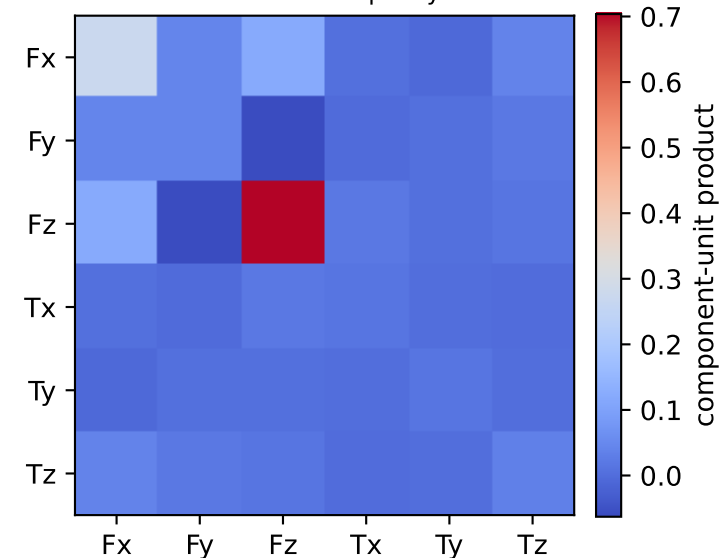
empirical total covariance



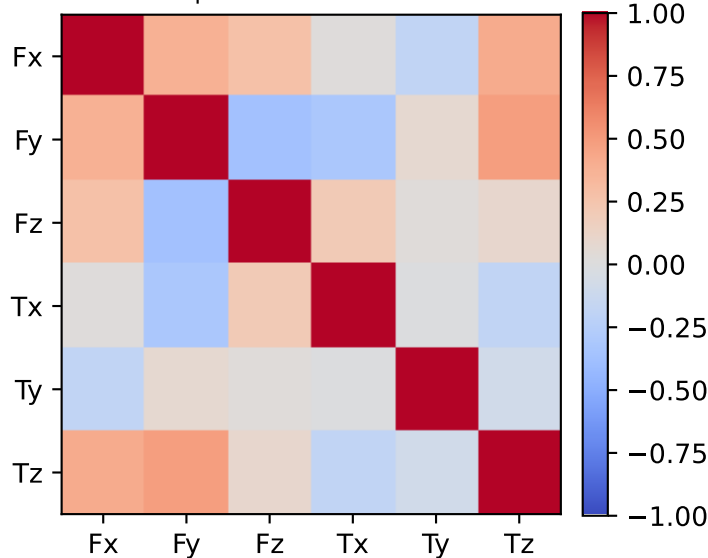
mean full-SG prediction covariance



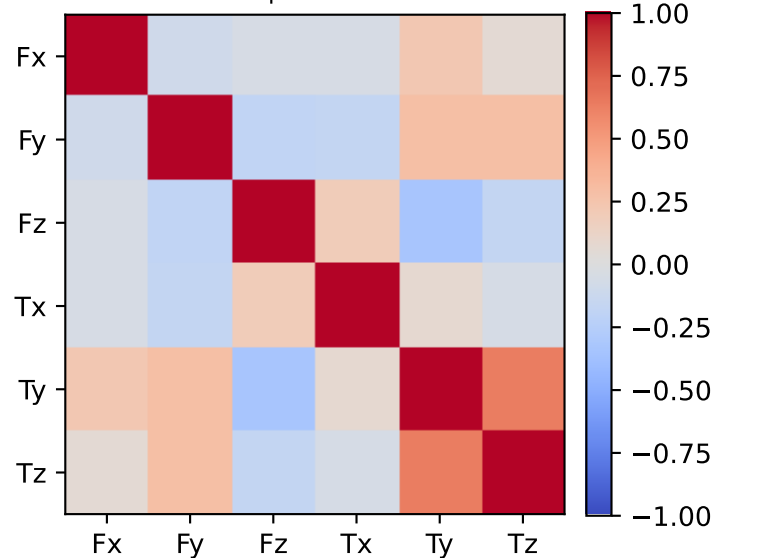
PSD excess model discrepancy covariance



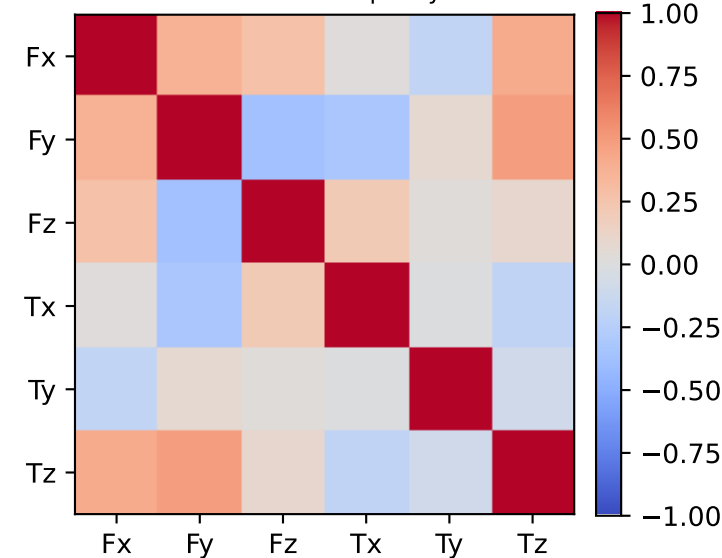
empirical total correlation



mean full-SG prediction correlation

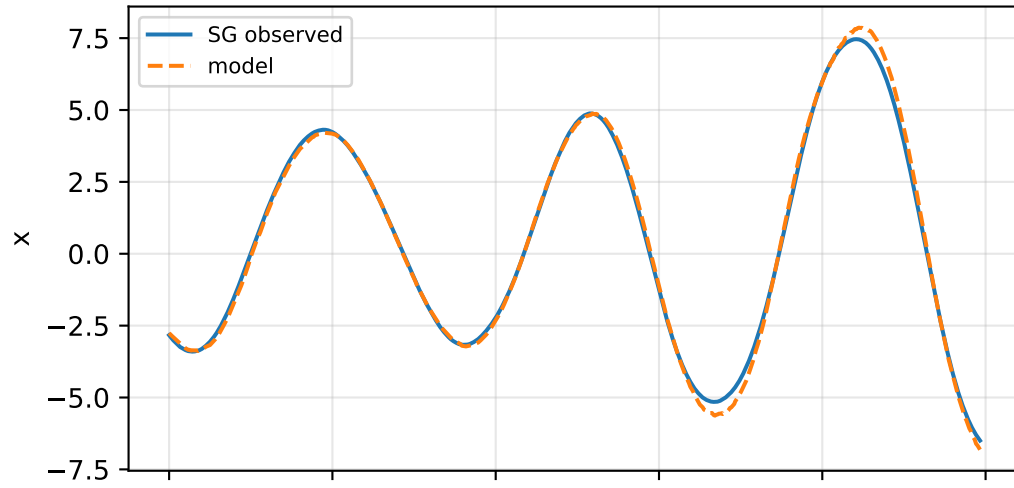


PSD excess model discrepancy correlation

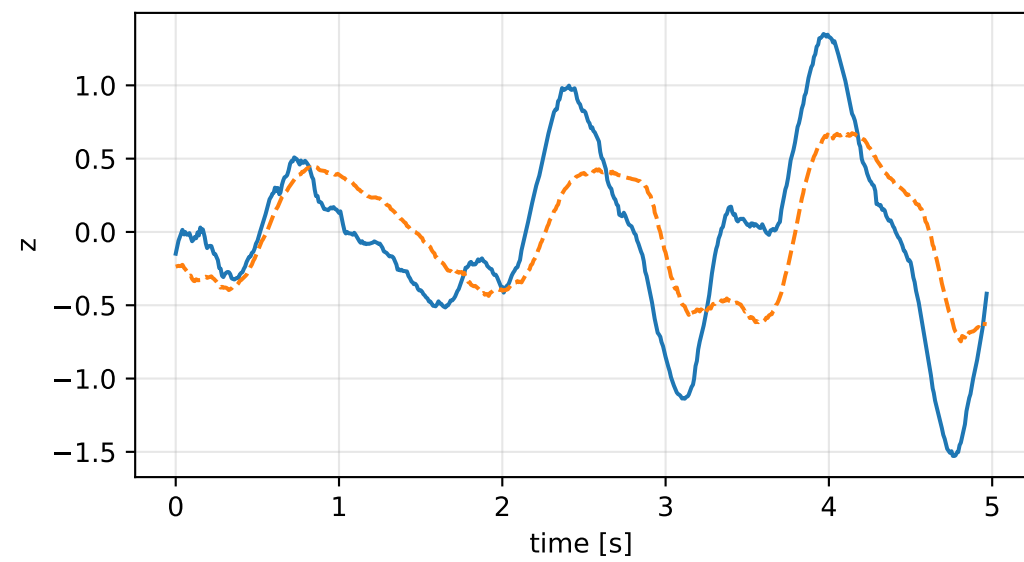
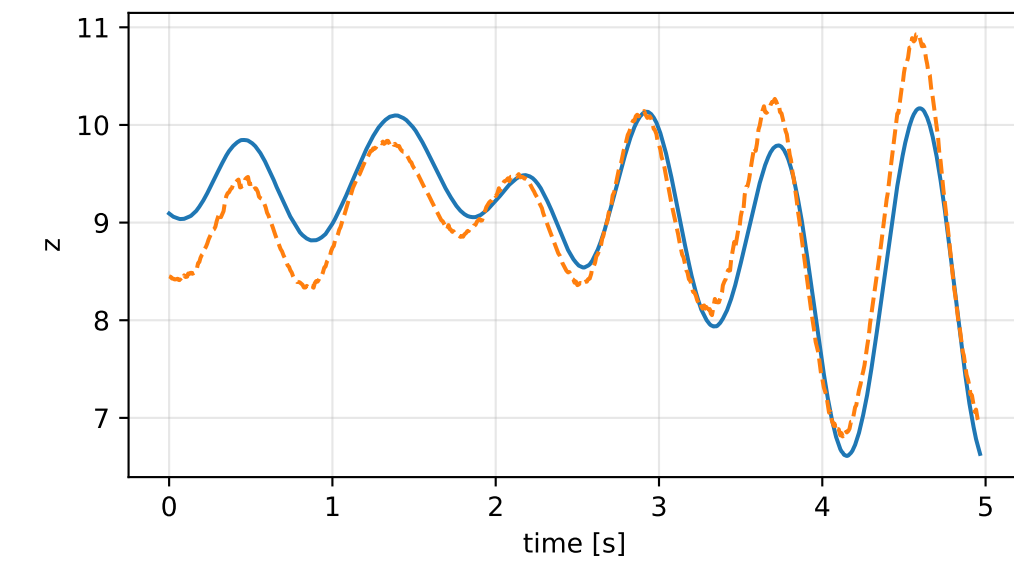
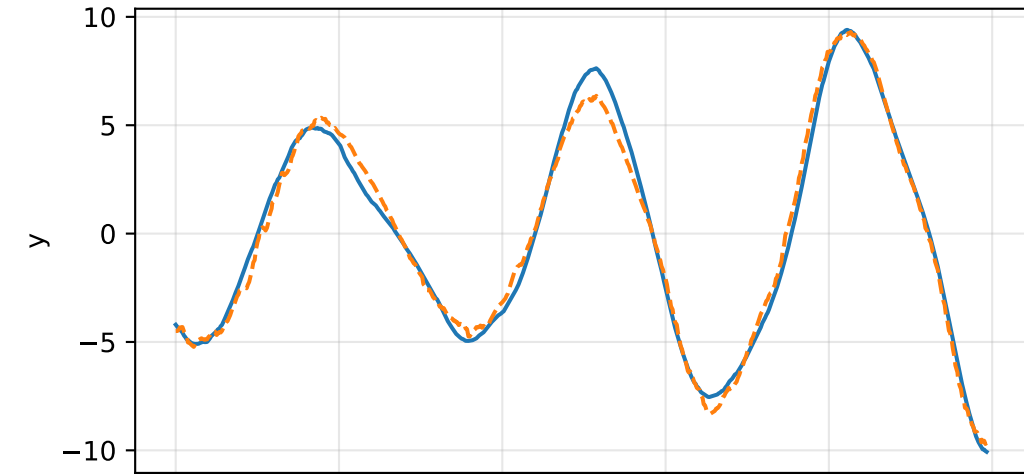
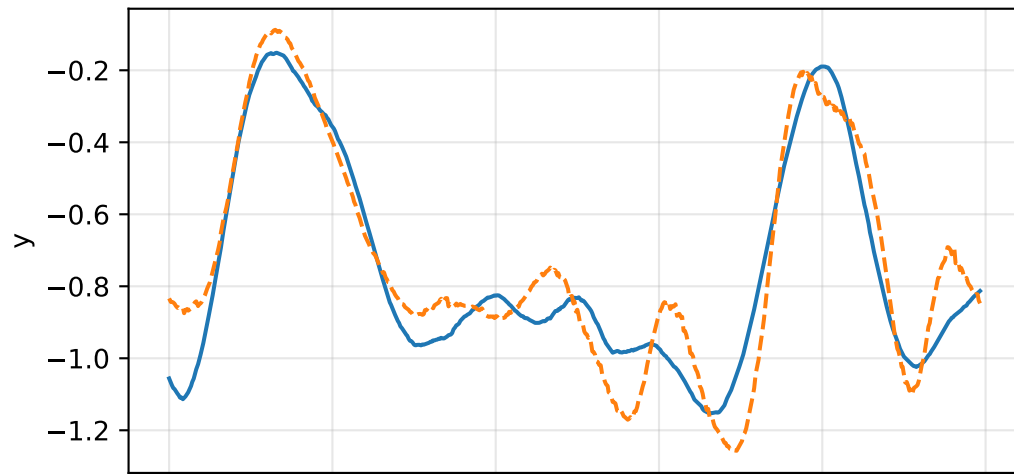
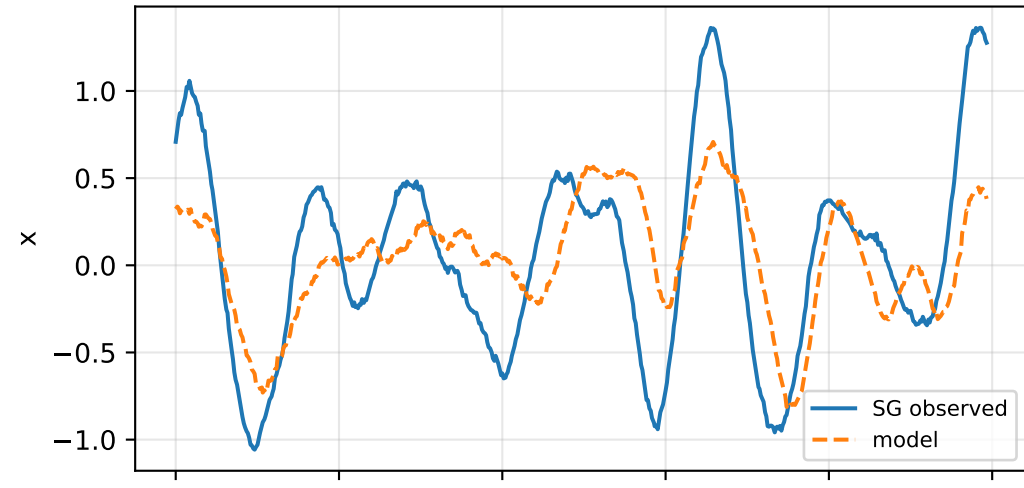


force_over_mass_rotor_2_nominal: acceleration residual objective

specific acceleration [m/s²]

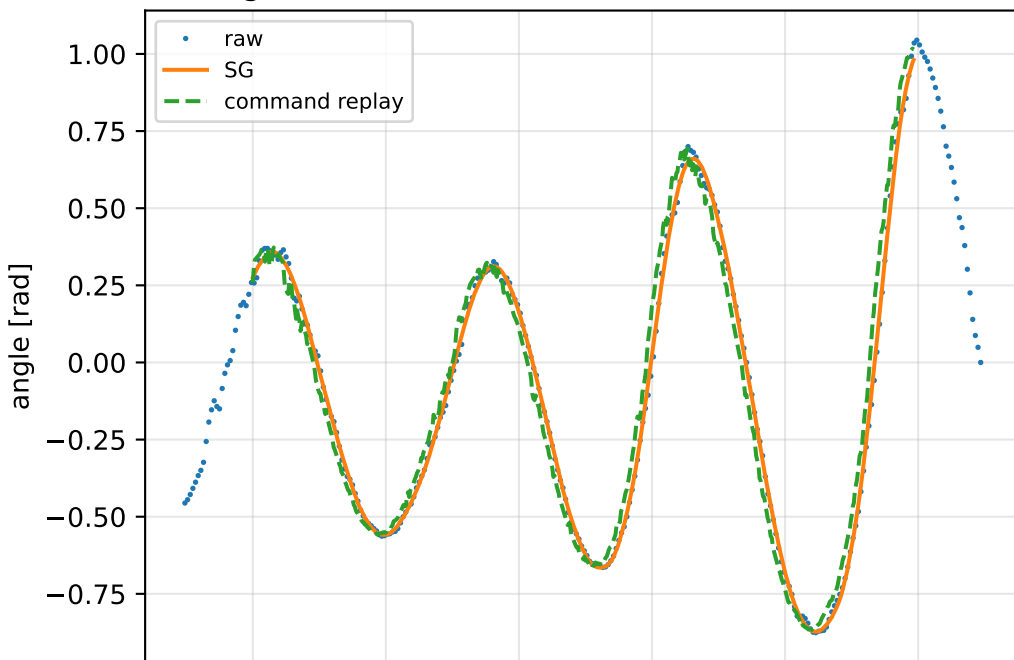


angular acceleration [rad/s²]

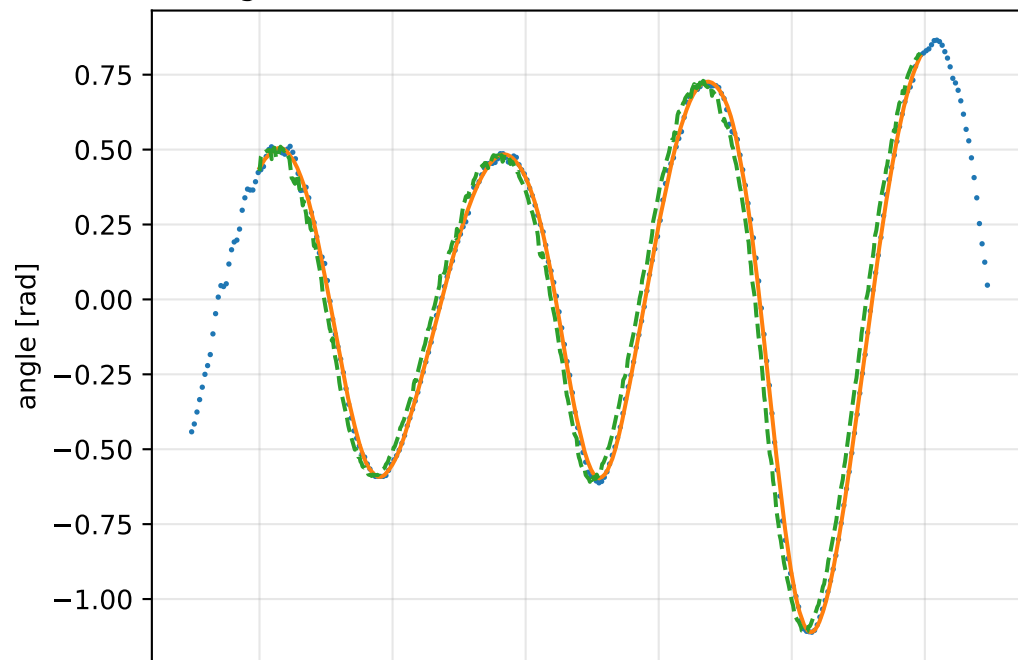


force_over_mass_rotor_2_nominal: gimbal measurement audit (objective=measured_sg)

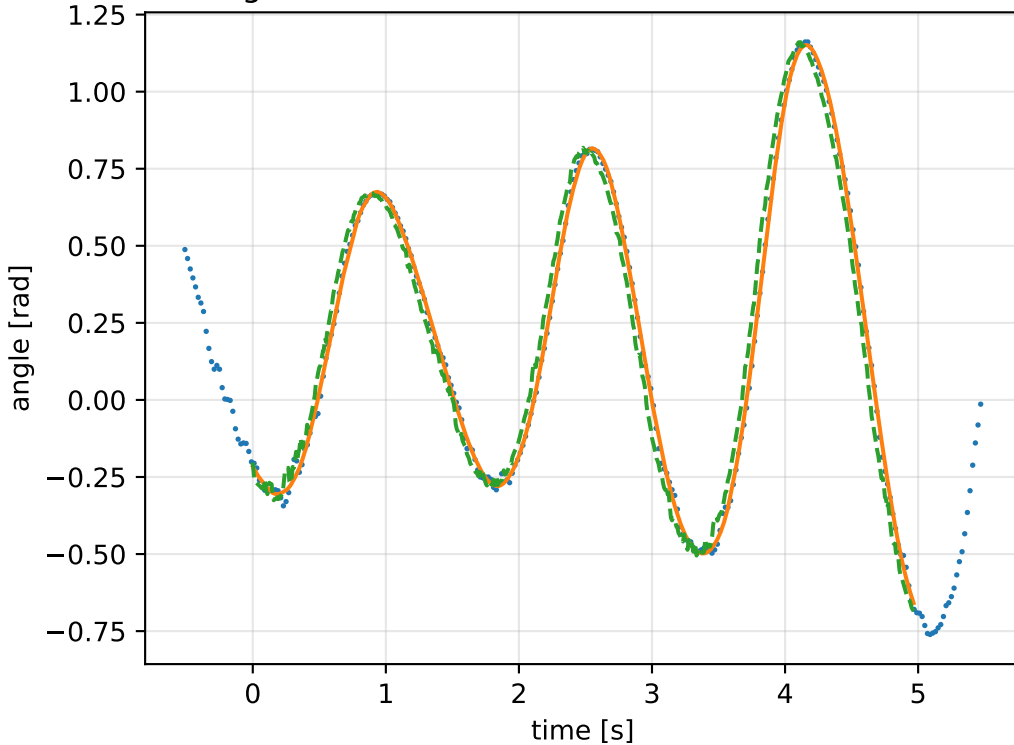
gimbal 1: RMSE=0.0759, max=0.2095 rad



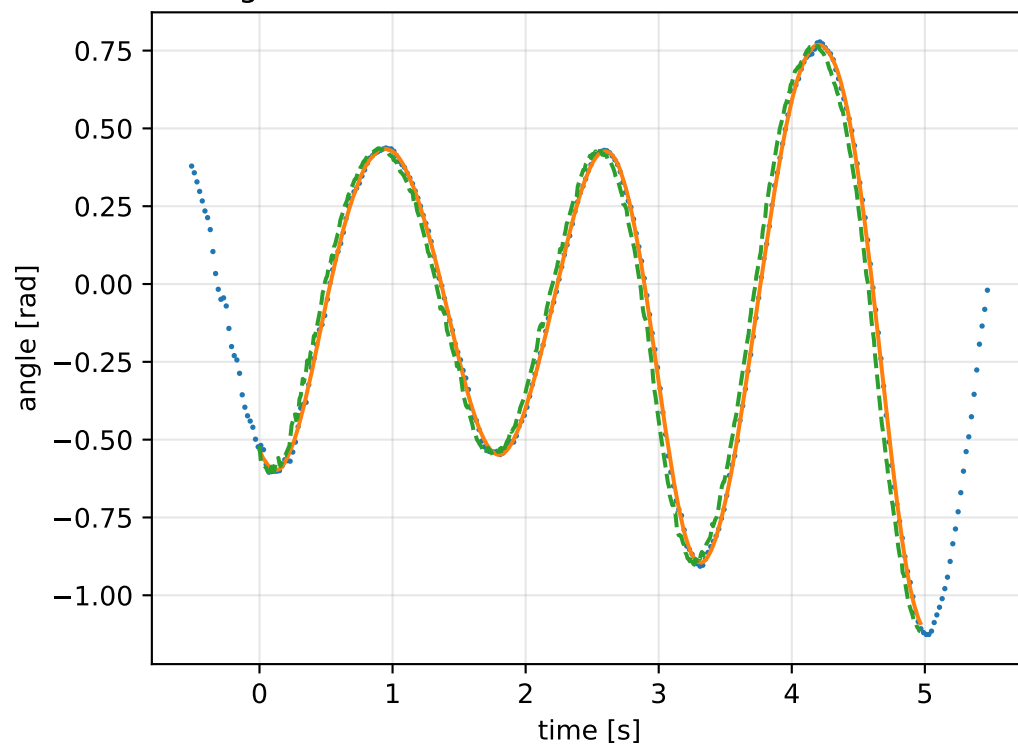
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

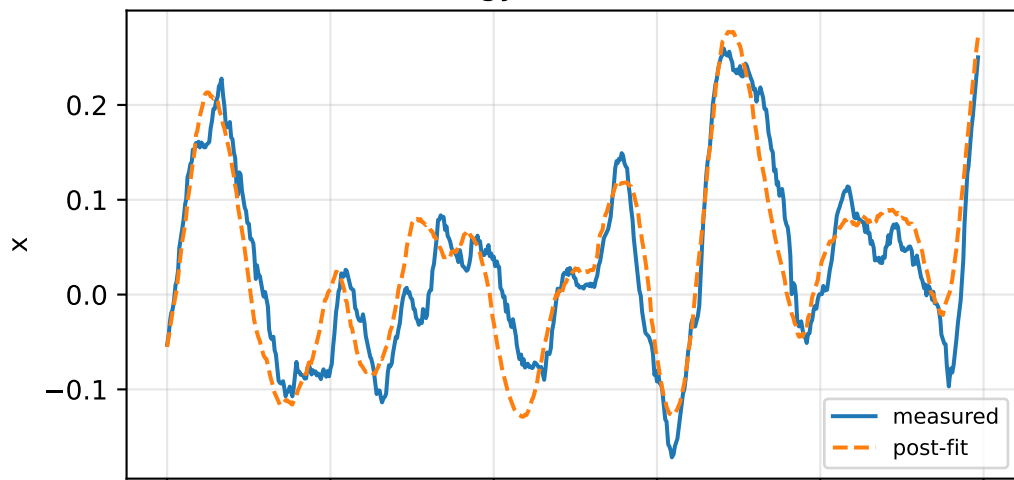


gimbal 4: RMSE=0.07131, max=0.1737 rad

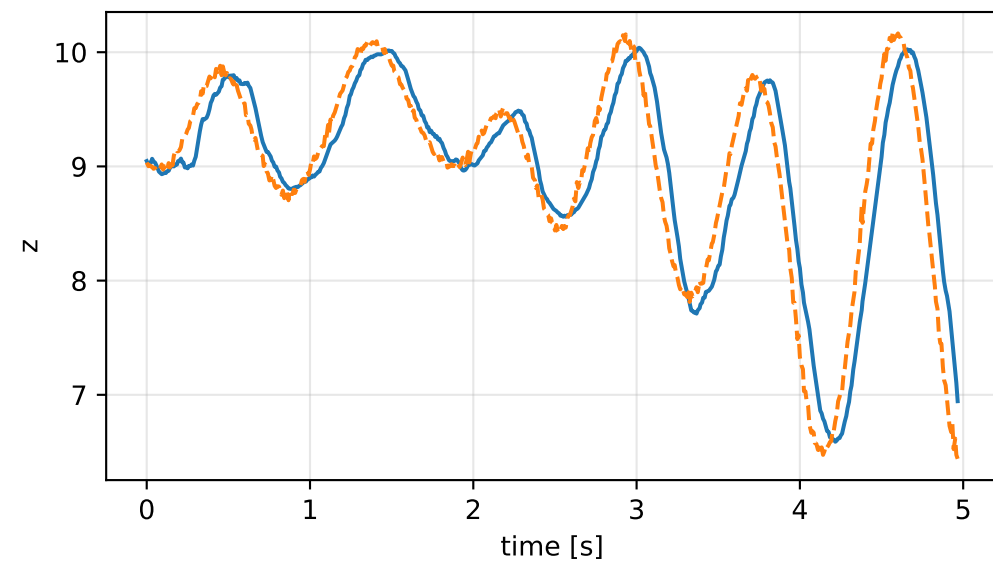
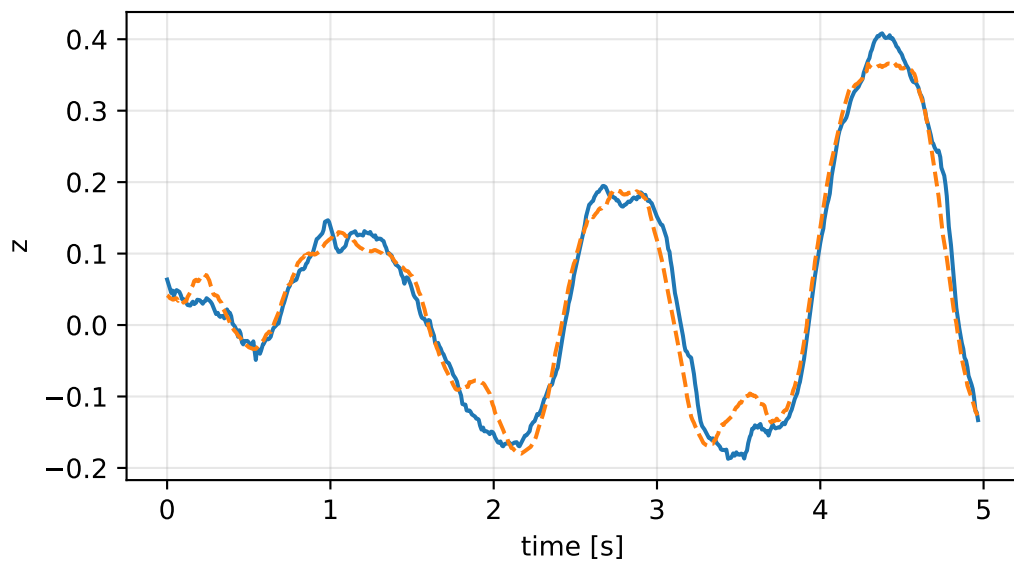
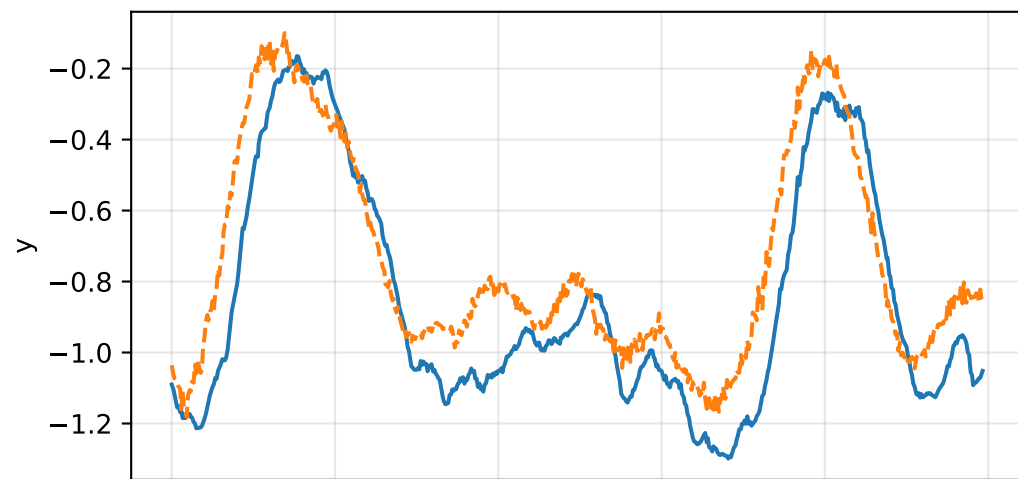
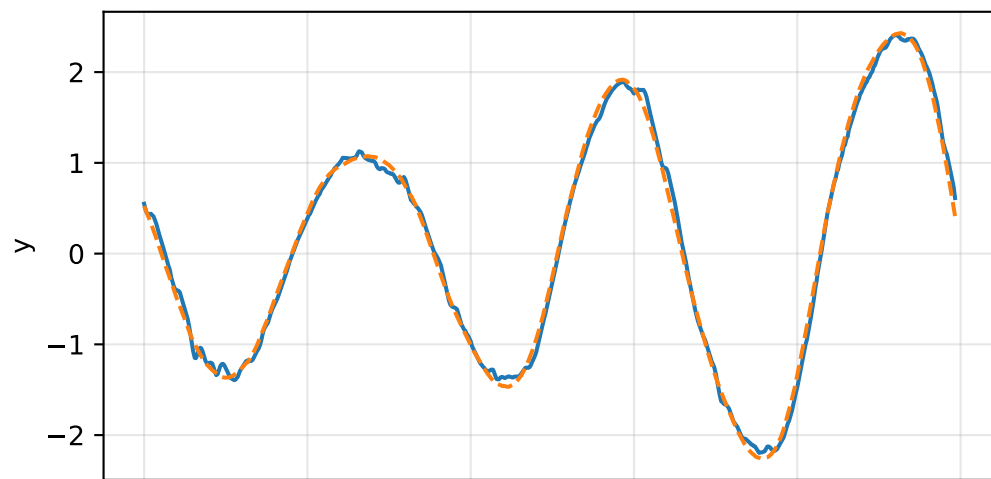
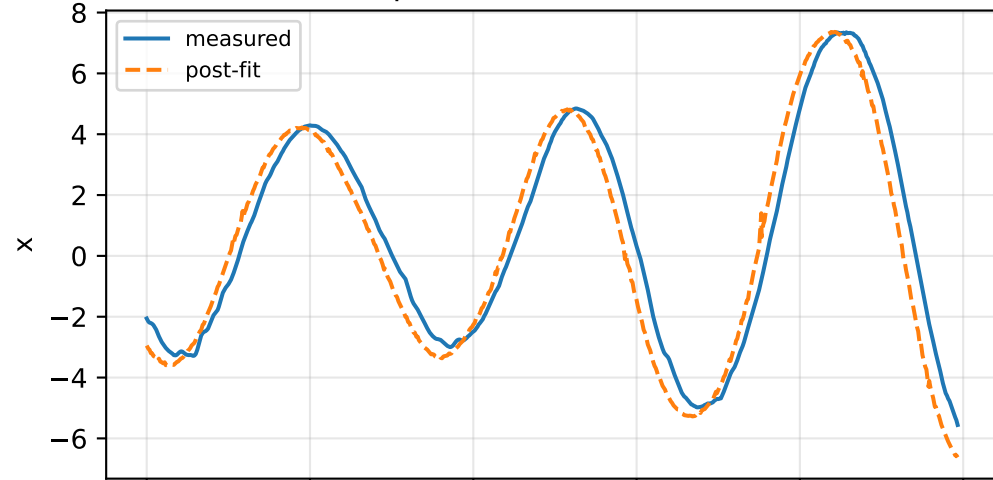


force_over_mass_rotor_2_nominal: IMU comparison (diagnostic only)

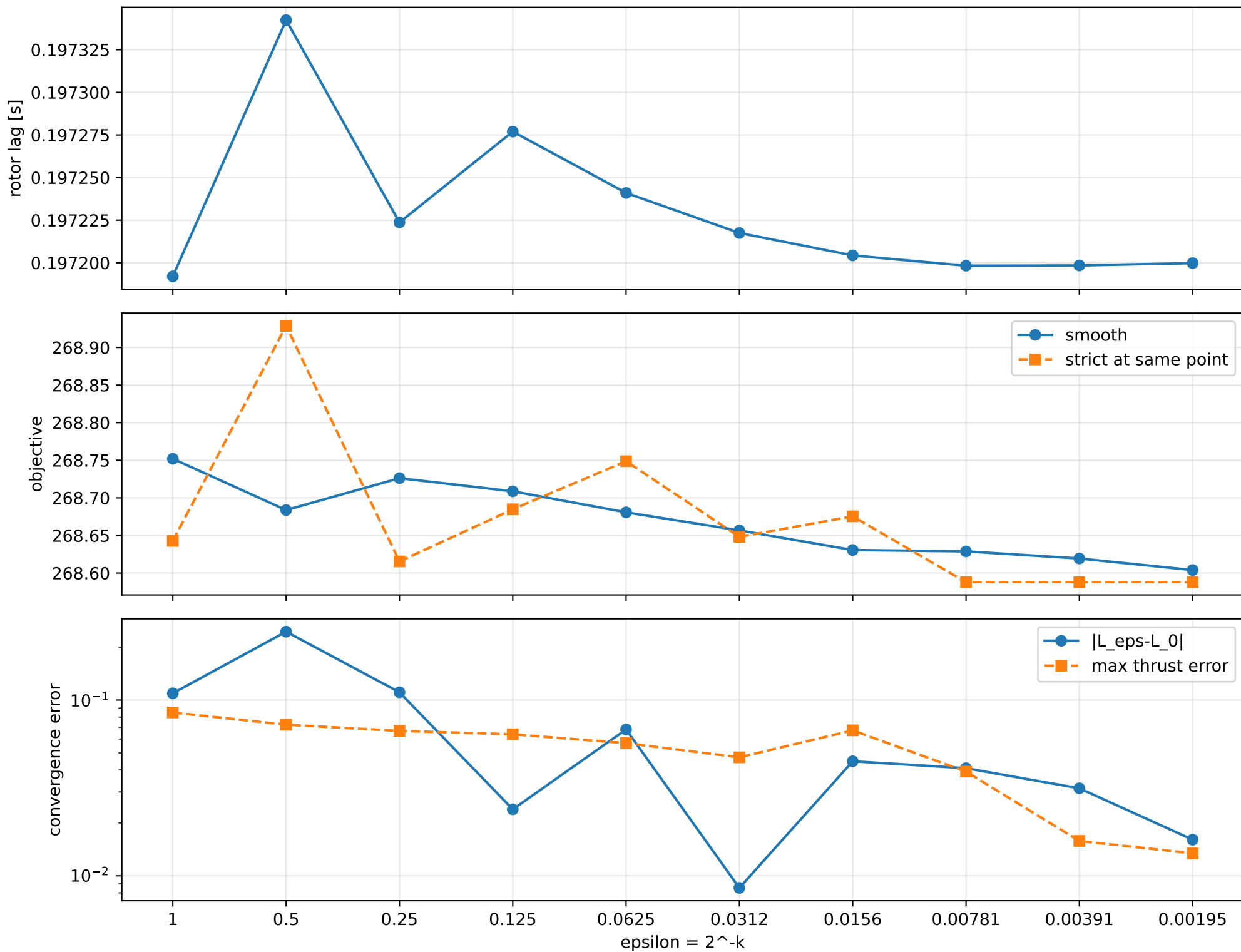
gyro [rad/s]



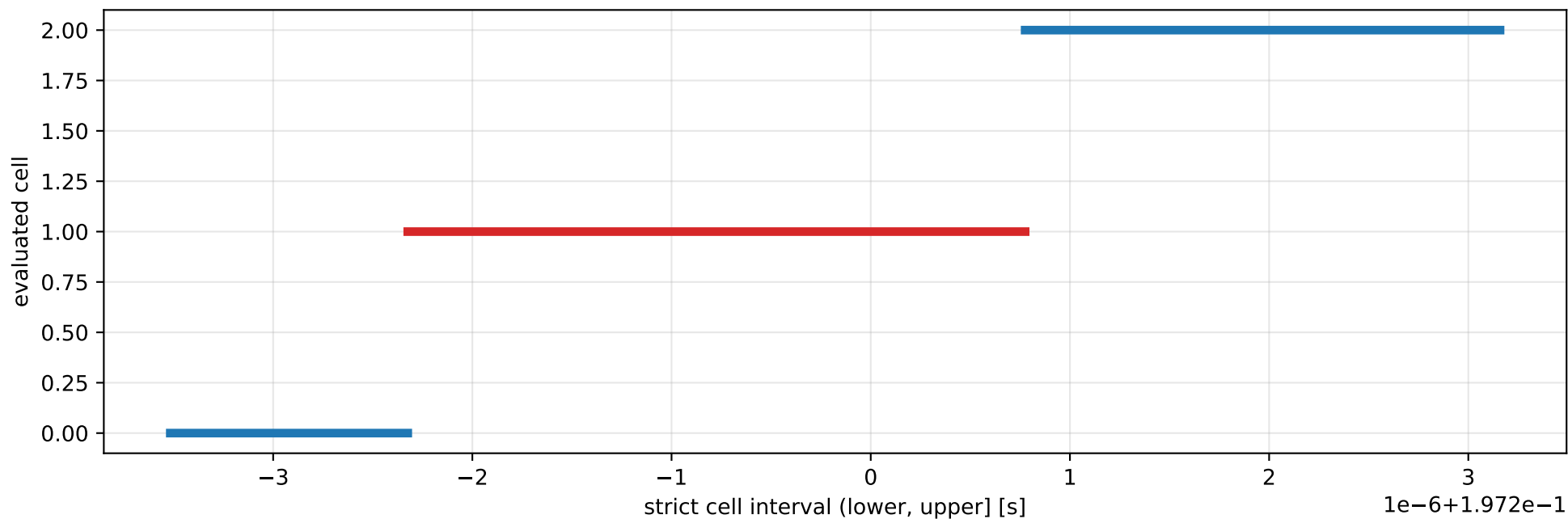
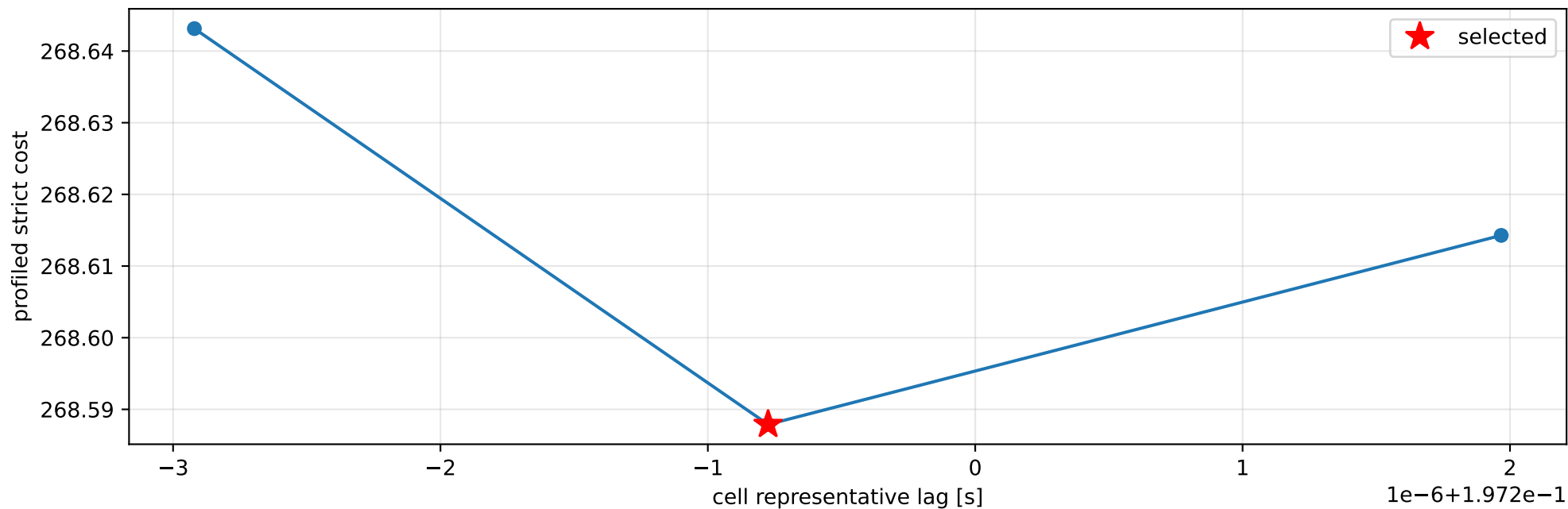
specific force [m/s^2]



force_over_mass_rotor_2_nominal: smooth-to-strict continuation



force_over_mass_rotor_2_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199804 s; data boundary=False

force_over_mass_rotor_2_nominal: parameter estimate

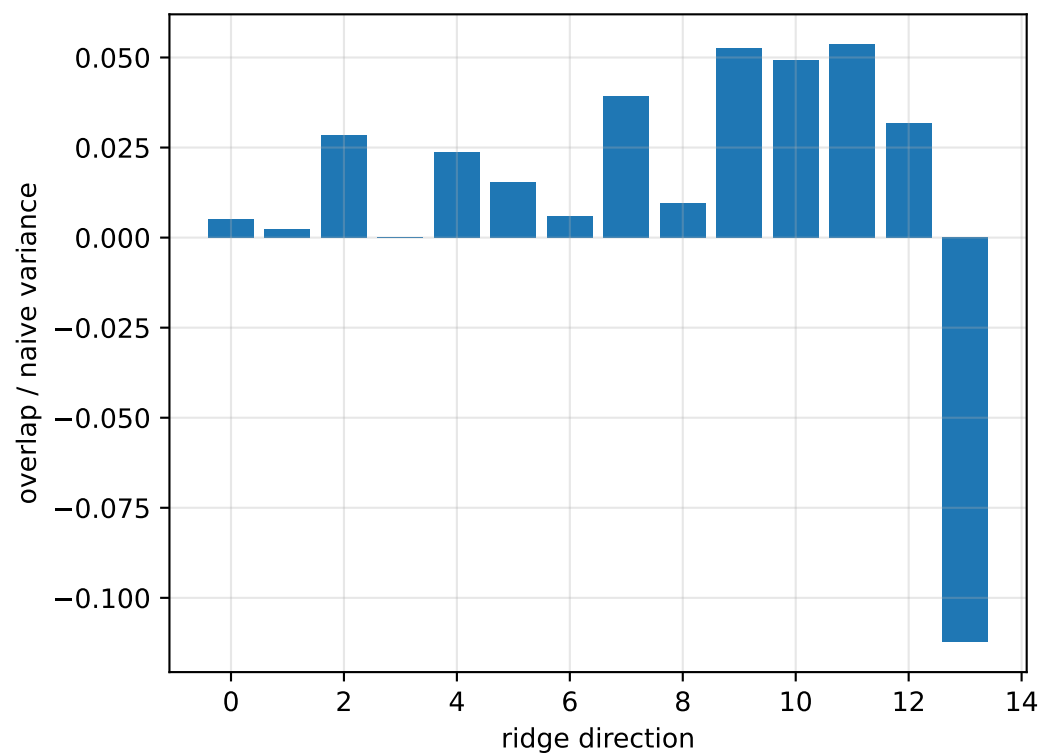
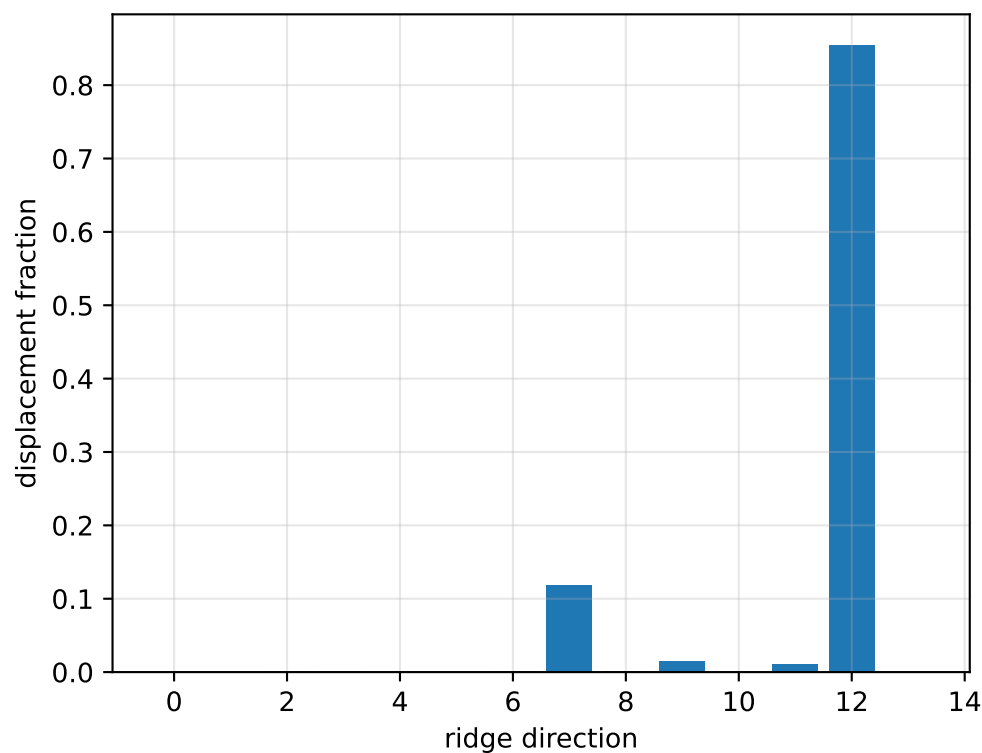
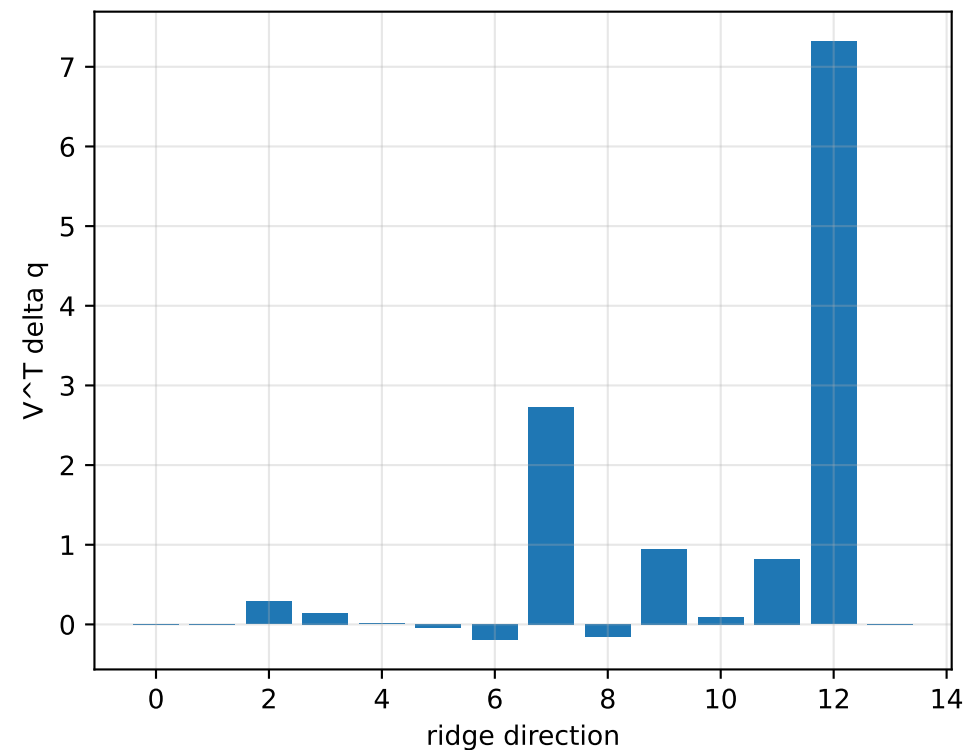
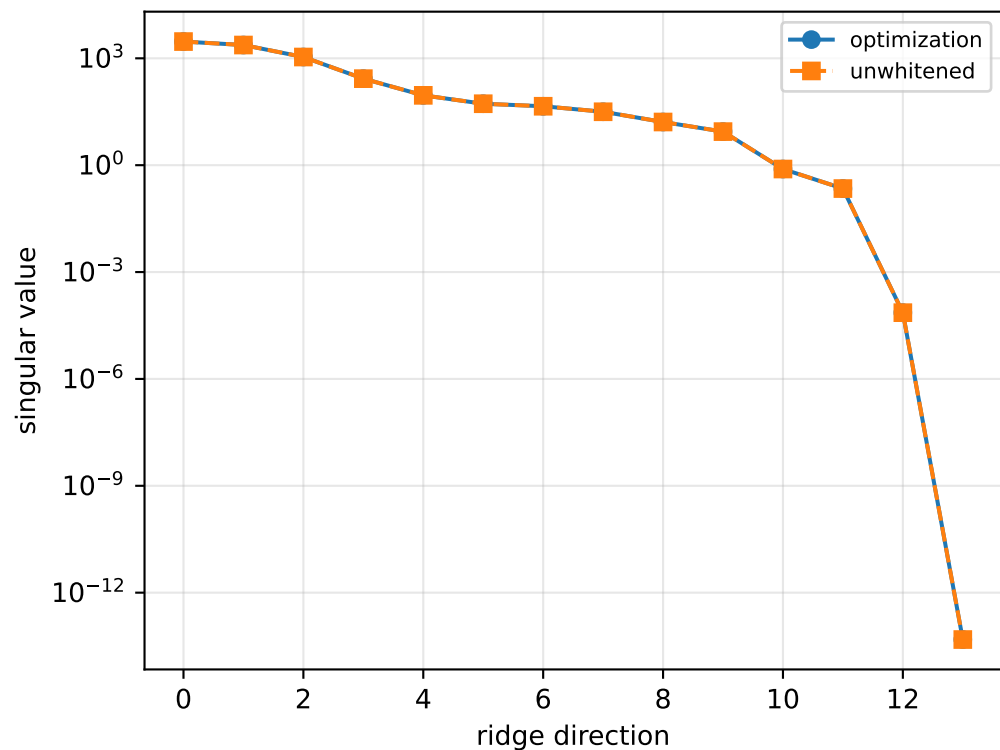
Case: force_over_mass_rotor_2_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.70517225
inertia [kg m²]:
[[3.72430696e-01 -2.38388968e-04 -6.17463162e-02]
 [-2.38388968e-04 2.87118303e-01 -1.90527236e-03]
 [-6.17463162e-02 -1.90527236e-03 6.31605611e-01]]
CoG body [m]: [0.01546384 -0.00276453 -0.0443783]
force effectiveness: [1.24740091 1.57559792 1.39031828 1.17532271]

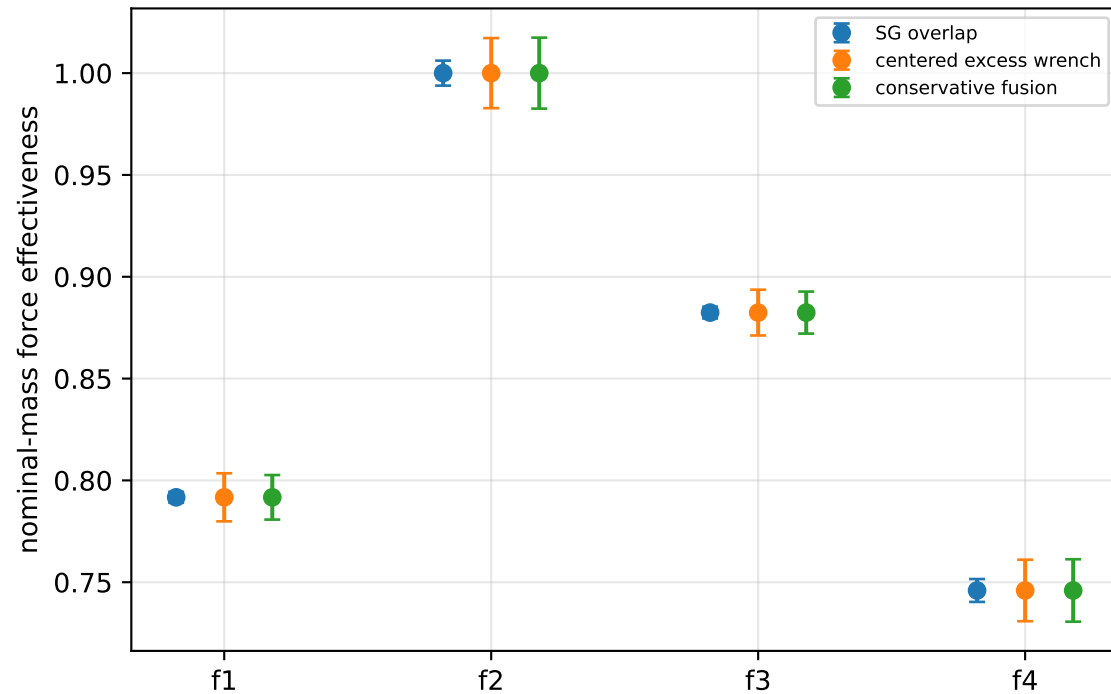
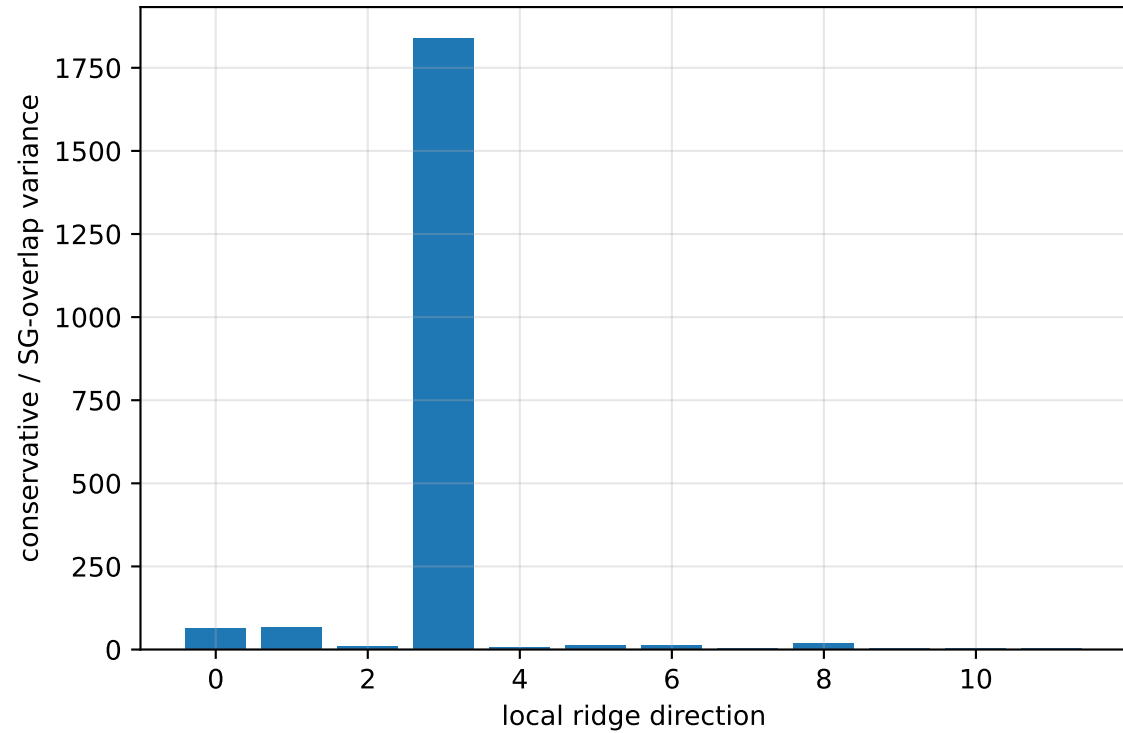
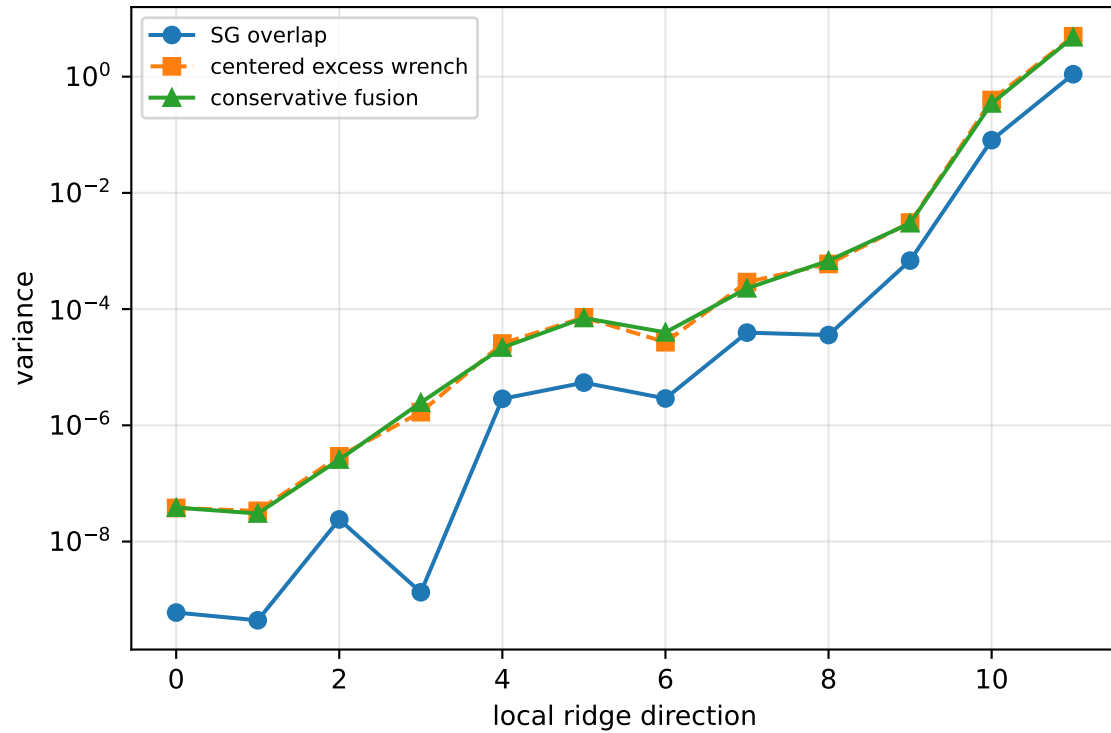
scale-free J/m [m²]:
[[1.00516433e-01 -6.43395104e-05 -1.66648976e-02]
 [-6.43395104e-05 7.74912160e-02 -5.14219644e-04]
 [-1.66648976e-02 -5.14219644e-04 1.70465924e-01]]
scale-free f/m: [0.33666476 0.42524283 0.37523715 0.31721136]

rotor lag cell: (0.197197676, 0.197200775] s
representative: 0.197199225 s

force_over_mass_rotor_2_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.03e-13)



force_over_mass_rotor_2_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.42$
wrench-to-acceleration recovery max error = $3.11\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

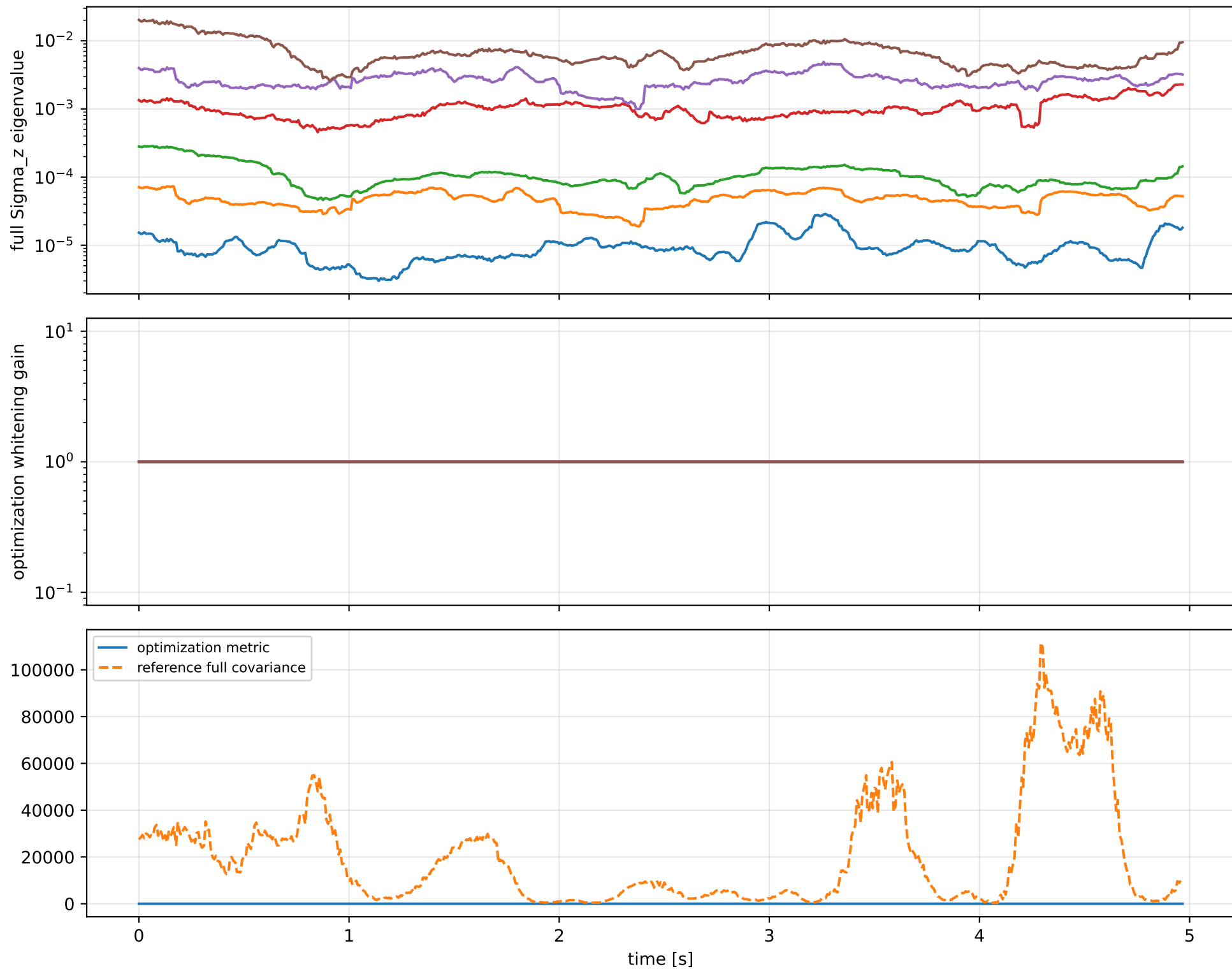
index 11: variance=4.771, ratio=4.33
 $\log_{\text{mass}}=-0.0459$; second_moment_diag_1=-0.416
second_moment_diag_2=-0.041; second_moment_diag_3=+0.686
second_moment_offdiag_12=+0.0829; second_moment_offdiag_13=-0.577
second_moment_offdiag_23=-0.0663; cog_x=+1.24e-05
cog_y=-0.000193; cog_z=+2.09e-05
 $\log_{\text{force_eff_1}}=-0.0455$; $\log_{\text{force_eff_2}}=-0.0449$
 $\log_{\text{force_eff_3}}=-0.0469$; $\log_{\text{force_eff_4}}=-0.0464$

index 10: variance=0.3463, ratio=4.286
 $\log_{\text{mass}}=-0.00476$; second_moment_diag_1=-0.0146
second_moment_diag_2=-0.0446; second_moment_diag_3=+0.0795
second_moment_offdiag_12=-0.886; second_moment_offdiag_13=-0.0689
second_moment_offdiag_23=+0.449; cog_x=-0.00135
cog_y=+0.00333; cog_z=-0.000324
 $\log_{\text{force_eff_1}}=-0.0122$; $\log_{\text{force_eff_2}}=-0.0219$
 $\log_{\text{force_eff_3}}=+0.00232$; $\log_{\text{force_eff_4}}=+0.0163$

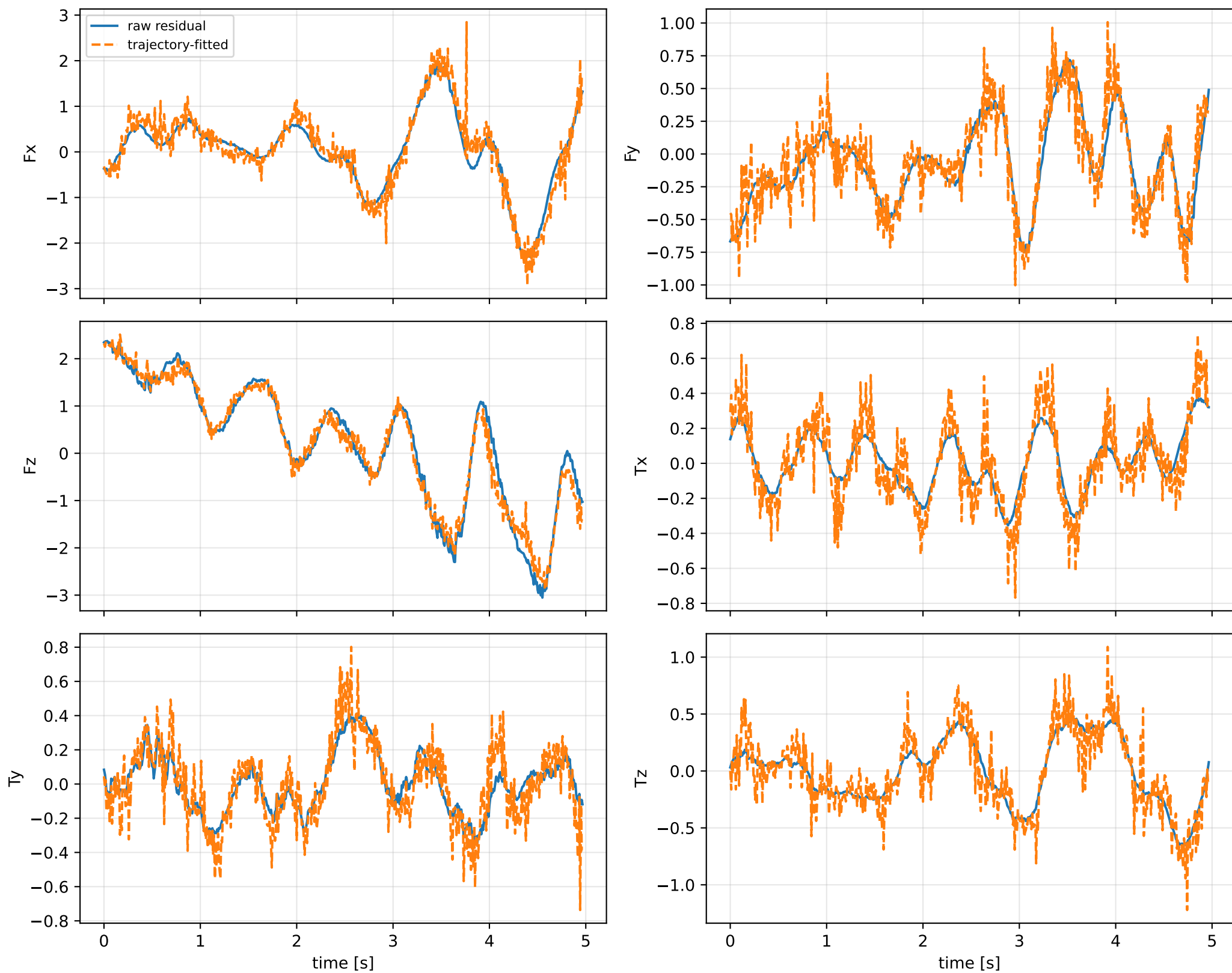
index 9: variance=0.003008, ratio=4.399
 $\log_{\text{mass}}=-0.151$; second_moment_diag_1=-0.0319
second_moment_diag_2=+0.902; second_moment_diag_3=-0.0804
second_moment_offdiag_12=-0.0278; second_moment_offdiag_13=-0.0824
second_moment_offdiag_23=+0.0393; cog_x=+0.0203
cog_y=-0.0125; cog_z=-0.00428
 $\log_{\text{force_eff_1}}=-0.186$; $\log_{\text{force_eff_2}}=-0.0346$
 $\log_{\text{force_eff_3}}=-0.0994$; $\log_{\text{force_eff_4}}=-0.318$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_2_nominal: optimization vs reference covariance



force_over_mass_rotor_2_nominal: wrench / closure (force max 1.15e-14, torque max 8.33e-16)



force_over_mass_rotor_2_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_2_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 4ad6b24989a0df886676644a499e15946486f27db58e2dfd259bd888c891980e

data objective: 268.587896265

prior objective: 2.38285866024e-08

total objective: 268.587896289

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_2_nominal (force_effectiveness_over_mass)

components: rotor_2

target: [0.42524282]

std: [1.e-05]

final: [0.42524283]

error: [2.18305229e-09]

standardized residual: [0.00021831]

factor objective: 2.38285866024e-08